Autocorrelation in Daily Stock Market Returns: Time-variation and Asymmetry

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Abstract

We examine the autocorrelation in daily U.S. stock market returns from 1897 to 2024, its time evolution and structural properties. We find substantial time variation in the autocorrelation – starting around zero before 1929, rising to be significantly positive and peaking in the 1970s, and turning persistently negative after 2000, with sharp drops during major crises. Despite these shifts, a consistent asymmetry emerges: return reversals are more likely following negative returns. Moreover, this asymmetry is robust across major global equity markets. Turning to the drivers of this autocorrelation, we find that a small number of extreme losses account for much of the negative autocorrelation, and volatility emerges as the most consistent predictor – particularly during market downturns. These findings provide useful insights on the different mechanisms underlying the observed daily return autocorrelation.

Keywords: Index Serial-Dependence, Market Downturns, Market Efficiency

JEL Classification: G12, G14

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1 Introduction

Influenced by the classic benchmark of the random walk hypothesis on stock prices, serial dependence in stock returns, especially at short horizons, has been viewed as an empirical measure of market imperfections. In this paper, we examine the serial dependence in daily stock index returns over a sample period spanning more than a century, from 1897 to 2024, providing new "facts" on its levels, time variation and empirical structure, which offer insights into the underlying mechanisms driving the autocorrelation dynamics.

We begin by documenting the substantial time variation in the serial dependence of daily stock index returns. Our analysis focuses on the Dow Jones Industrial Average (DJIA), a widely recognized benchmark for the U.S. equity market, with data available dating back to 1897. Using a 5-year rolling window from 1897 to 2024, we estimate the serial dependence coefficient at the end of each month m, denoted by ρ_m, following the regression:

$$r_{t+1} = a_m + \rho_m \cdot r_t + \epsilon_t,$$

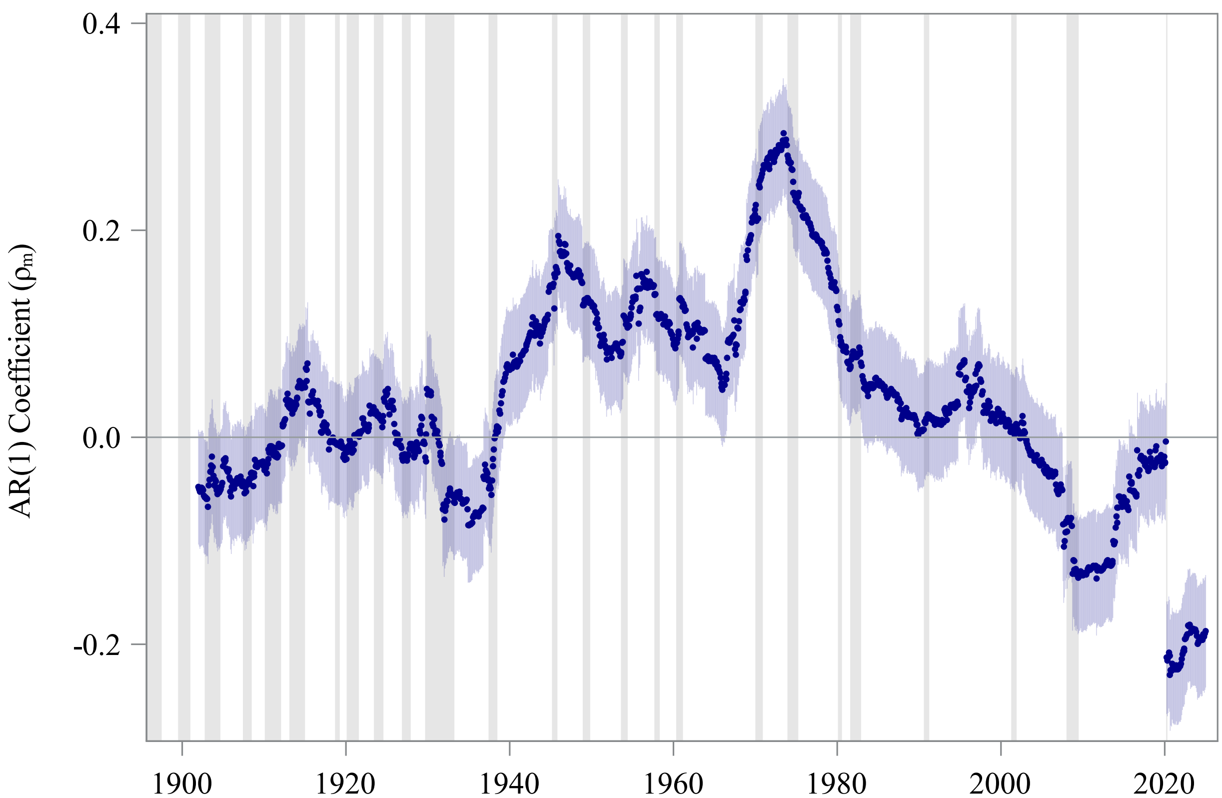
using daily returns $\{r_t\}$ during the window.[1]

As shown in Figure 1, the estimated serial dependence ρ_m of DJIA daily returns exhibits pronounced variation over the full sample period. Prior to the onset of the Great Depression in 1929, serial dependence remains modest in magnitude and statistically insignificant, with ρ_m fluctuating within a narrow range around zero. During the Great Depression (1929–1939), serial dependence declines sharply, turning significantly negative and reaching a local minimum of around –0.09 in the early 1930s. In the aftermath of the Depression, ρ_m increases steadily, turning positive in the 1940s and remaining predominantly positive for several decades. It peaks at 0.29 in the early 1970s – the highest level of positive serial dependence observed in the entire sample – before declining gradually in the following decades. Starting around 2000, serial dependence weakens further and eventually turns negative. During the Global Financial Crisis (2008–2009), ρ_m drops sharply, reaching a local trough of approximately –0.14. It then begins a slow recovery, approaching zero by February 2020, just prior to the onset of the COVID-19 pandemic. In the immediate aftermath of the pandemic shock,

[1] Throughout this paper, we use the terms "serial dependence" and "autocorrelation" interchangeably for ρ_m.

serial dependence declines abruptly once more, reaching –0.23 – the lowest value observed in the sample. While ρ_m rebounds modestly in the subsequent years, it remains in negative territory through the end of 2024.

Figure 1: Serial Dependence of Dow Jones Index Returns. This figure plots the serial dependence of daily returns for the Dow Jones Industrial Average Index, represented by the coefficient ρ_m following the equation $r_{t+1} = a_m + \rho_m \cdot r_t + \epsilon_t$, at each month-end m from 1897 to 2024 based on a 5-year rolling window of daily index returns r_t. Gray-shaded regions represent NBER-designated recessions, while blue-shaded areas represent the 95% confidence intervals for the ρ_m estimates.



In contrast to the large time variation in equity index serial dependence, we identify a robust asymmetry: the serial dependence between day t and day $t + 1$ is significantly more negative following negative returns r_t. As illustrated in Figure 2, the serial dependence conditional on negative r_t consistently lies below that for positive r_t across most of the sample period. Over the full sample, the estimated serial dependence is -0.08 following negative returns and -0.02 following positive returns, resulting in a statistically significant difference of -0.06 (t-statistic = -2.32). This asymmetric pattern is persistent throughout the sample and becomes particularly pronounced during specific sub-periods. For instance, the difference in serial dependence reaches lows of -0.24 in the pre-1929 era, -0.34 during the 1960s, and -0.27 near the turn of the millennium.[2]

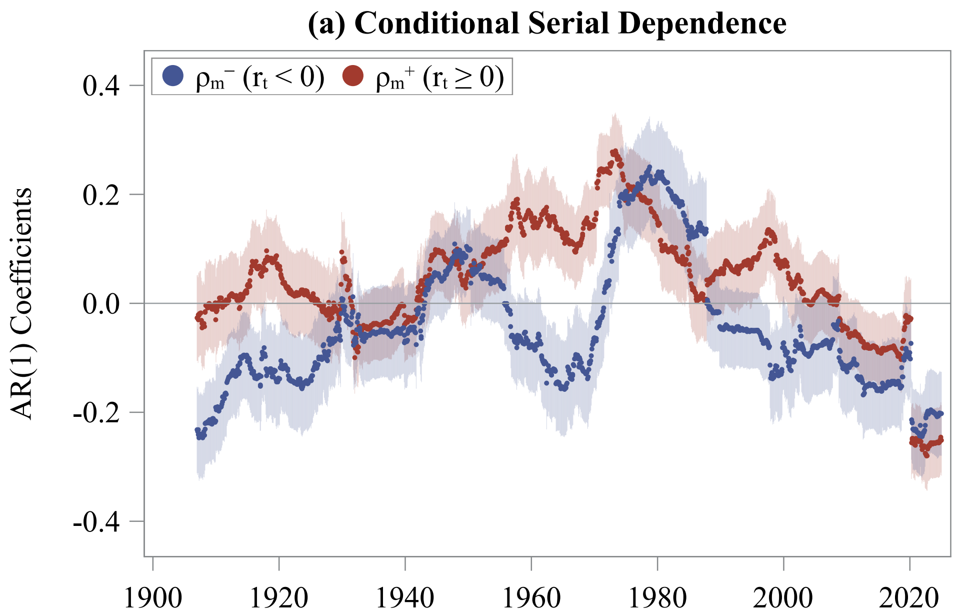
Further analysis reveals another dimension of asymmetry: extreme negative returns exert a disproportionately strong influence on serial dependence, while extreme positive returns have minimal impact. To isolate this effect, we exclude the most extreme return days incrementally and recompute serial dependence. When observations with r_t less than -8%, -4%, and -2% are removed, the serial dependence following negative returns weakens from -0.08 to -0.04, -0.02, and eventually becomes positive to 0.06. This represents a changes of -0.14 (t-statistic = -4.30) between the full-sample estimate and the truncated sample with r_t between $(-2\%, 0\%)$. Notably, this large change is driven by a small subset of the data – just 12, 182, and 1,062 days, corresponding to only 0.1%, 0.6%, and 3.4% of the full sample, respectively. In contrast, excluding the most extreme positive return days produces little change: the difference in serial dependence between all days with positive r_t and those between $[0\%, 2\%)$ is -0.04 and statistically insignificant. These results suggest that return reversals in equity indices are largely driven by left-tail events.

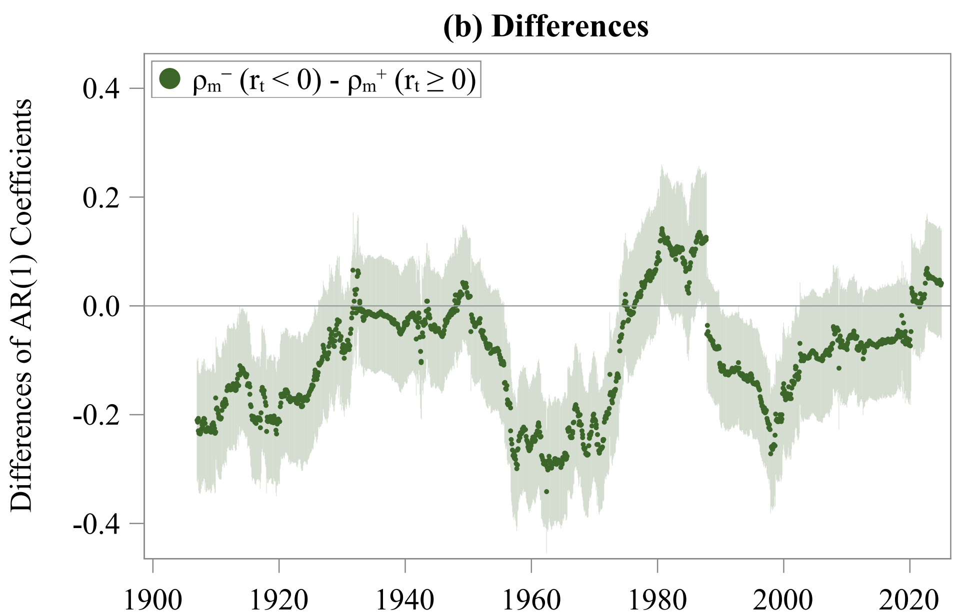
Extending the above analysis to global equity indices reveals similar asymmetric patterns, suggesting that this phenomenon is pervasive across international markets. We examine eight additional major indices from Japan, Hong Kong, France, the U.K., Canada, Germany, Australia, and an additional U.S. benchmark (S&P 500). In every market studied, we observe a consistent result: serial dependence is more negative following negative returns than after positive ones. The average difference in autocorrelation between days following negative and positive return days ranges from -0.07 for France's CAC 40 Index to -0.18 for Japan's TOPIX Index. Outside the U.S., the asymmetry is particularly pronounced in Japan, where the difference in autocorrelation reached as much as -0.25 in the 1960s and remained sizable at -0.11 in the 2000s. These findings highlight a robust and globally consistent pattern: serial dependence tends to be more negative during market downturns across global equity markets and different time periods.[3]

[2]Based on the sign of the serial dependence, we also show that a trading strategy exploiting this signal can generate returns with significantly higher Sharpe ratios than the market benchmark. These results are available upon request.

[3]In unreported results, we also examine serial dependence using S&P 500 index futures data, which

Figure 2: Asymmetric Serial Dependence of Dow Jones Index Daily Returns. This figure shows the asymmetric serial dependence of daily returns for the Dow Jones Industrial Average Index from January 1897 to December 2024. The coefficients $\rho_m^+(r_t \geq 0)$ and $\rho_m^-(r_t < 0)$ are estimated monthly using a 10-year rolling window of daily returns, based on the equation $r_{t+1} = a_m + \rho_m \cdot r_t + \epsilon_t, \quad \text{for } t \in [m - 10\text{y}, m]$, conditional on the sign of r_t. Specifically, $\rho_m^+(r_t \geq 0)$ is estimated using observations where $r_t \geq 0$, and $\rho_m^-(r_t < 0)$ is estimated using observations where $r_t < 0$. Subfigure (a) plots the time series of the two conditional serial dependence estimates. Subfigure (b) displays their difference, $\rho_m^-(r_t < 0) - \rho_m^+(r_t \geq 0)$. Shaded regions denote 95% confidence intervals of the estimated serial dependence.





Lastly, we investigate how various market conditions influence index serial dependence. Among all factors considered, volatility emerges as the most significant and consistent driver of autocorrelation in both U.S. and global equity markets. Higher volatility is associated with more negative serial dependence: for example, a one standard deviation increase in volatility changes the serial dependence of the Dow Jones by -0.03 (t-stat = –4.56), and by an average of -0.01 across global indices. Importantly, this volatility effect is also asymmetric, appearing primarily on negative-return days. In the U.S., a one standard deviation increase in volatility changes serial dependence by -0.07 (t-stat = –4.35) when conditioned on negative return days, compared to an insignificant impact for positive return days. Similar asymmetries are observed across international markets. In contrast, other market variables exhibit weaker and less consistent effects. In the U.S., serial dependence tends to be more negative when inflation is low or the term spread is wide, but these patterns are not robust globally. Overall, our findings underscore volatility – particularly during downturns – as a key driver of return reversals and the asymmetry in index serial dependence.

Our study contributes to the extensive literature on short-horizon non-zero stock market return predictability, a topic of longstanding interest in finance since Kendall and Hill (1953). While various mechanisms have been proposed in the literature to explain the serial dependence in equity index returns, the substantial variation in its magnitude over time suggests that no simple explanation can fully capture its underlying dynamics.

For instance, the non-synchronous trading channel (Lo and MacKinlay 1988; Atchison et al. 1987; Lo and MacKinlay 1990; Boudoukh et al. 1994; Ahn et al. 2002) offers a plausible explanation for the positive serial dependence observed between 1941 and 2000. However, it does not account for the negative serial dependence seen both before 1941 and after 2000. Similarly, the channel of sluggish response to information and partial price adjustments (Brennan et al. 1993; Badrinath et al. 1995; McQueen et al. 1996; Chordia and Swaminathan 2000; Moskowitz et al. 2012) implies positive serial dependence of index returns, but can not explain episodes of strong negative serial dependence, such as during sharp market corrections or high-volatility environments, where price reversals are more

extends back to 1982. The results exhibit similar time variation and asymmetric patterns as those observed for the Dow Jones and S&P 500 indexes.

prevalent.

The observed negative serial dependence is consistent with a liquidity-based explanation, in which significant price reversals reflect compensation to liquidity providers (Ho and Stoll 1981; Grossman and Miller 1988; Campbell, Grossman, and Wang 1993; Chordia, Roll, and Subrahmanyam 2002; Duffie 2010). While this channel may not account for the strong positive serial dependence observed between the 1940s and 1990s, it aligns well with the robust asymmetric patterns seen across global markets – particularly because collateral and funding constraints tend to tighten more following market declines. Additionally, the significant effect of volatility on serial dependence is consistent with the liquidity channel, as higher volatility amplifies the inventory risk borne by liquidity providers. But the liquidity mechanism does easily explain the positive autocorrelation from 1940 to 1990.

Another possible mechanism is time-varying risk and liquidity premiums, driven by the short horizon variations in risk, including volatility and beyond, and the risk tolerance of marginal investors in the market, similar to the model in (Campbell, Grossman, and Wang 1993).

More recently, attention has shifted to the emergence of negative serial dependence in post-2000 periods. Baltussen et al. (2019) attributed the post-2000 decline in serial dependence to demand pressures driven by the rise of index-linked investment products. While this theory helps explain part of the downward trend immediately after 2000, it appears inconsistent with the near-zero autocorrelation observed before the COVID-19 pandemic in 2020, as well as the sharp and sustained negative autocorrelation that followed. The indexing channel also could not explain the negative serial dependence observed during the Great Depression.

The long horizon changes in daily return autocorrelation indicate that many mechanisms may be at work at the same time and their relative importance may change over time as the structure and technology of the market evolves. More work is needed to capture these mechanisms in a coherent framework and their time evolution.

Two contemporaneous studies (Lewellen 2022; Bogousslavsky et al. 2024) also examine and document the significant negative and asymmetric serial dependence in the U.S. equity index after 2000. Our work differs in several respects. First, we use an extended sample spanning 1897 to 2024, allowing us to highlight substantial time variation in serial dependence. We show that the asymmetric effect persists even in earlier decades when unconditional serial dependence was largely positive. Furthermore, we identify that a small number of extreme negative returns account for much of the observed asymmetry. Finally, our study is the first to document that similar patterns of time-varying and asymmetric serial dependence are also present in global equity markets.

The rest of the paper is organized as follows. Section 2 presents stylized facts on serial dependence in equity index returns, highlighting its asymmetry during market downturns, the disproportionate impact of extreme negative return days, and its relationship with volatility. Section 3 explores potential mechanisms driving these patterns. Section 4 concludes.

2 Stylized Facts

2.1 Asymmetric Serial Dependence of the Dow Jones Index Returns

We first show that the serial dependence of Dow Jones Index returns is asymmetric. Specifically, days with negative returns tend to be followed by stronger negative correlations with next-day returns compared to days with positive returns, suggesting a greater tendency for reversals following market declines.

To quantify this effect, we estimate the serial dependence coefficients separately using the regression $r_{t+1} = a_m + \rho_m \cdot r_t + \epsilon_t$ at each month-end m from 1897 to 2024, conditional on the sign of r_t. The estimates are computed using a 10-year rolling window of daily index returns, so that the conditional serial dependence estimates are based on a comparable number of observations to those used in the unconditional analysis. We denote the coefficient estimated from observations with $r_t \geq 0$ as ρ_m^+ and that from observations with $r_t < 0$ as ρ_m^-. As illustrated in Figure 2, although the two conditional serial dependence measures generally follow similar trends over time, the coefficient conditional on negative returns, $\rho_m^-(r_t < 0)$ (blue line), consistently lies below its counterpart for positive returns, $\rho_m^+(r_t \geq 0)$ (red line), across most of the sample period. This persistent gap results in predominantly negative values for the difference $\rho_m^-(r_t < 0) - \rho_m^+(r_t \geq 0)$, shown as the green line in the bottom panel of Figure 2.

Substantial negative differences in the two conditional serial dependence are evident in several key historical periods. For instance, the gap reaches approximately -0.24 during the pre-1929 era, -0.34 in the 1950s–1960s, and -0.27 right before the turn of 2000s. These episodes reflect periods when the market exhibited a pronounced tendency to reverse following daily losses. In contrast, during other intervals – such as 1931 to 1950 and 2020 to 2024 – the difference is close to zero and not statistically significant, suggesting a more symmetric response to positive and negative returns. The only notable exception occurs briefly in the early 1980s, when the difference temporarily turns positive and becomes marginally significant.

These findings underscore the asymmetric nature of serial dependence in equity index returns. When conditioned on the sign of current returns, the market consistently exhibits a greater tendency for short-term reversals following negative return days. Moreover, this asymmetry in serial dependence following negative returns is robust across time and does not depend on whether the unconditional serial dependence is positive or negative. In fact, we observe the largest gap between the conditional serial dependence during the 1950s and 1960s, when the unconditional serial dependence is largely positive.

2.2 Effects of Extreme Returns on the Serial Dependence

Next, we examine the role of extreme returns in driving the asymmetric serial dependence observed in U.S. equity index returns. To evaluate these effects, we incrementally exclude the most extreme daily return observations and re-estimate the serial dependence. Table 1 reports the estimated serial dependence coefficients ρ from the regression $r_{t+1} = a + \rho \cdot r_t + \epsilon_t$, as the range of current returns r_t is incrementally restricted – from $(-\infty, 0\%)$ to $(-2\%, 0\%)$ for negative returns, and from $[0\%, \infty)$ to $[0\%, 2\%)$ for positive returns. Panel A presents results for the full sample of DJIA returns, while Panels B through E provide estimates for successive 30-year subperiods.

Excluding extreme negative return days substantially weakens the negative serial dependence observed in index returns. In the full sample spanning 1897 to 2024, the conditional serial dependence following negative return days is estimated at -0.08 and is statistically significant. When observations with r_t below -8%, -4%, and -2% are sequentially excluded, the estimated autocorrelation rises from -0.08 to -0.04, -0.02, and ultimately turns posi-

Table 1: Effects of Extreme Returns on the Serial Dependence of Dow Jones Index Returns. This table reports the serial dependence of daily returns for the Dow Jones Industrial Average Index. Serial-dependence ρ are estimated from the regression $r_{t+1} = a + \rho \cdot r_t + \epsilon_t$ for various ranges of r_t. The left panel reports the serial-dependence after excluding extreme negative returns, while the right panel reports then serial-dependence after removing extreme positive returns. Results are presented for the full sample (Panel A) as well as for different sub-samples (Panels B to E). T-statistics, calculated based on Newey-West standard errors, are shown in square brackets.

	Negative return days ($r_t < 0$)					Positive return days ($r_t \geq 0$)				
	(1)	(2)	(3)	(4)	(1) - (4)	(5)	(6)	(7)	(8)	(5) - (8)
	(-∞, 0%)	(-8%, 0%)	(-4%, 0%)	(-2%, 0%)		[0%, ∞)	[0%, 8%)	[0%, 4%)	[0, 2%)	
Panel A: Full Sample 1897.01-2024.12										
ρ	-0.08***	-0.04*	-0.02	0.06**	-0.14***	-0.02	-0.02	0.00	0.02	-0.04
	[-3.56]	[-1.82]	[-1.34]	[2.48]	[-4.30]	[-0.84]	[-0.77]	[0.19]	[0.90]	[-1.22]
Nobs	14982	14970	14800	13920		17061	17043	16921	16103	
(%)	46.8	46.7	46.2	43.4		53.2	53.2	52.8	50.3	
Panel B: Subsample 1897.01-1929.12										
ρ	-0.11**	-0.10**	-0.08**	-0.04	-0.08	0.03	-0.01	0.01	0.04	-0.01
	[-2.43]	[-2.25]	[-2.40]	[-0.82]	[-1.23]	[0.96]	[-0.39]	[0.27]	[1.31]	[-0.20]
Nobs	3786	3781	3742	3460		4394	4392	4372	4141	
(%)	46.3	46.2	45.8	42.3		53.7	53.7	53.5	50.6	
Panel C: Subsample 1930.01-1959.12										
ρ	-0.06	-0.01	-0.05	0.07	-0.13*	-0.05	-0.05	-0.04	-0.02	-0.03
	[-1.22]	[-0.38]	[-1.31]	[1.27]	[-1.76]	[-1.23]	[-1.18]	[-0.94]	[-0.36]	[-0.42]
Nobs	3511	3508	3422	3152		3994	3983	3922	3705	
(%)	46.8	46.7	45.6	42.0		53.2	53.1	52.3	49.4	
Panel D: Subsample 1960.01-1989.12										
ρ	0.01	0.14	0.13***	0.16***	-0.15**	0.10***	0.13***	0.08***	0.07**	0.03
	[0.21]	[1.54]	[3.90]	[4.69]	[-2.41]	[3.79]	[3.16]	[3.34]	[2.13]	[0.78]
Nobs	3629	3627	3620	3529		3914	3913	3901	3753	
(%)	48.1	48.1	48.0	46.8		51.9	51.9	51.7	49.8	
Panel E: Subsample 1990.01-2024.12										
ρ	-0.14***	-0.10***	-0.02	0.04	-0.18***	-0.10*	-0.07*	-0.01	-0.01	-0.08
	[-3.18]	[-2.88]	[-0.60]	[0.88]	[-2.94]	[-1.70]	[-1.71]	[-0.42]	[-0.43]	[-1.24]
Nobs	4056	4054	4016	3779		4759	4755	4726	4504	
(%)	46.0	46.0	45.6	42.9		54.0	53.9	53.6	51.1	

tive at 0.06. This represents a significant increase of 0.14 between the full-sample estimate and the truncated sample with $r_t \in (-2\%, 0\%)$, with a t-statistic of -4.30.

Importantly, this significant change is driven by a very small subset of observations: only 12, 182, and 1,062 days fall below the -8%, -4%, and -2% thresholds, accounting for just 0.1%, 0.6%, and 3.4% of the full sample, respectively. The disproportionate impact of these extreme left-tail returns is consistent across time. In four rolling 30-year subperiods, excluding returns below -2% leads to increases in conditional serial dependence of 0.08, 0.13, 0.15, and 0.18, with corresponding t-statistics of -1.23, -1.76, -2.41, and -2.94. The last two subperiods show statistically significant changes.

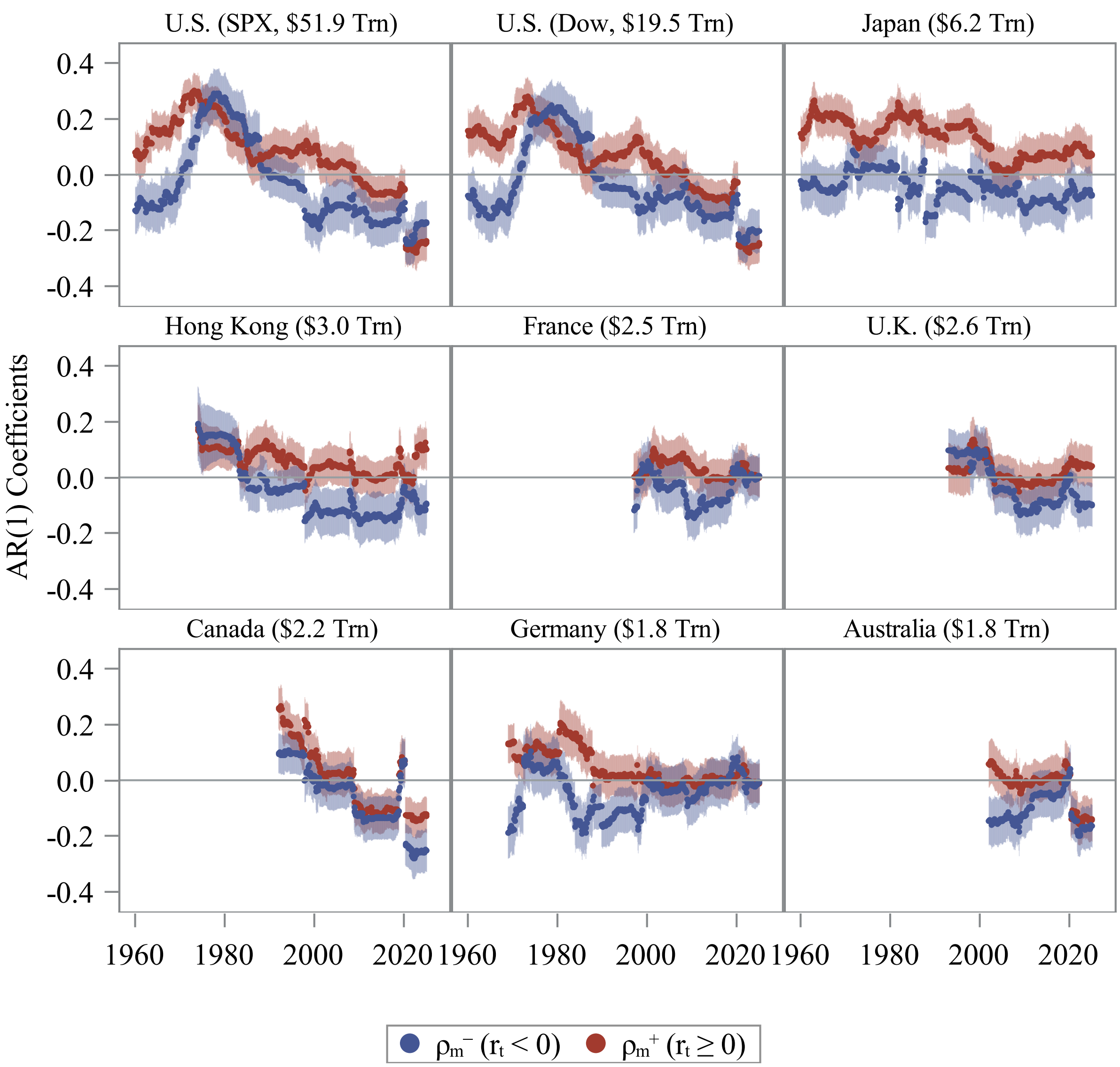
By contrast, excluding the most extreme positive return days yields small effect. The difference in serial dependence between the full sample of positive r_t and the truncated sample with $r_t \in [0\%, 2\%)$ is just -0.04, and statistically insignificant. Similarly, across the four 30-year subperiods, excluding returns above 2% results in changes of -0.01, -0.03, 0.03, and -0.08, with corresponding t-statistics of -0.20, -0.42, 0.78, and -1.24 – none of which are statistically significant.

Overall, these results reveal another dimension of asymmetry: extreme negative returns exert a disproportionately large influence on serial dependence, while extreme positive returns have negligible impact. This suggests that short-term reversals in equity indexes are primarily driven by left-tail events.

2.3 Evidences on Global Equity Indices

We then extend our analysis to a set of major global equity indices, selecting eight additional markets based on their market capitalization and international importance. These include the S&P 500 Index (U.S.), Tokyo Stock Price Index (TPX, Japan), Hang Seng Index (Hong Kong), CAC 40 Index (France), FTSE 100 Index (U.K.), S&P/TSX 60 Index (Canada), DAX Index (Germany), and S&P/ASX 200 Index (Australia). For each index, we estimate conditional serial dependence using the same methodology applied to the Dow Jones Industrial Average. Figure 3 presents the conditional serial dependence estimates for these indices. Due to data limitations, all series begin in 1960. For comparison, the Dow Jones Index results from 1960 onward are also included in the figure.

Figure 3: Serial Dependence of Global Equity Indexes. This figure shows the conditional serial dependence of global equity market indices from January 1960 to December 2024, including the S&P 500 Index and Dow Jones Index in the U.S., the Tokyo Stock Price Index in Japan, the Hang Seng Index in Hong Kong, the CAC 40 Index in France, the FTSE 100 Index in the U.K., the S&P/TSX 60 Index in Canada, the DAX Index in Germany, and the S&P/ASX 200 Index in Australia. The coefficients $\rho_m^+(r_t \geq 0)$ and $\rho_m^-(r_t < 0)$ are estimated monthly using a 10-year rolling window of daily returns, based on the equation $r_{t+1} = a_m + \rho_m \cdot r_t + \epsilon_t$, for $t \in [m - 10\text{y}, , m]$, conditional on the sign of r_t. Specifically, the red line represents $\rho_m^+(r_t \geq 0)$, estimated using observations where $r_t \geq 0$, and the blue line represents $\rho_m^-(r_t < 0)$, estimated using observations where $r_t < 0$. The shaded areas represent 95% confidence intervals for the $\rho_m^+(r_t \geq 0)$ and $\rho_m^-(r_t < 0)$ estimates. Index daily return and market cap data are from Bloomberg. The indices are ranked based on market cap as of December 31, 2024.



The time-series patterns observed across these global indices closely mirror those of the Dow Jones Index. In particular, the S&P 500 exhibits an almost identical trend, further validating our findings within the U.S. market. Across all global indices, there is a gradual decline in conditional serial dependence beginning around the early 2000s. Consistent with the results of Baltussen, van Bekkum, and Da (2019), several global markets – including Canada's S&P/TSX 60 and Australia's S&P/ASX 200 – exhibit negative serial dependence post-2000, echoing the shift observed in U.S. equity markets.[4]

More importantly, we find robust evidence of asymmetric serial dependence across all global equity markets. For each of the nine indices, the conditional serial dependence on positive return days (red lines) consistently exceeds that on negative return days (blue lines) over most of the sample period. The average difference in serial dependence ranges from -0.07 for France's CAC 40 Index to -0.18 for Japan's TPX Index. This asymmetry is especially pronounced in the markets of Japan, Hong Kong, and the U.S. For example, in Japan, the conditional serial dependence following positive return days ranges from 0.002 to 0.27, remaining consistently above zero throughout the 1960–2024 period. In contrast, the coefficient following negative return days remains below zero for most of the sample. The magnitude of the serial dependence asymmetry in Japan ranges ranges from -0.05 in December 1971 to -0.39 in September 1981.

Taken together, these findings indicate that global equity markets exhibit a similar asymmetric pattern in return dynamics: short-term reversals are more likely following market declines than gains. This consistent asymmetry across countries and time suggests the presence of a common, possibly structural or behavioral, mechanism underlying global return autocorrelation.

2.4 The Impact of Market Variables on Index Serial Dependence

Lastly, we examine how different market conditions influence equity index return serial dependence. To ensure robustness and avoid the smoothing effects of rolling windows, we

[4] Figure 3 focuses on conditional serial dependence and highlights the asymmetric patterns on positive and negative return days. In additional (unreported) results, we also examine unconditional serial dependence for these indices, which reveal similar trends – a transition from positive to negative autocorrelation around the early 2000s.

estimate serial dependence using non-overlapping quarters. Specifically, for each quarter q, we estimate the serial dependence coefficient ρ_q using the daily return of the Dow Jones index within quarter q.

We consider three categories of explanatory variables. The first group captures equity market conditions, such as equity index quarterly return and volatility. The second group comprises macroeconomic indicators that reflect broader economic conditions – such as inflation, the 3-month interest rate, the term spread (the yield difference between long- and short-term government bonds), and the NBER recession indicator. The third group includes measures of intermediary constraints and market mispricing, including the capital ratio of financial intermediaries from He, Kelly, and Manela (2017) and the investor sentiment index from Baker and Wurgler (2006).

To examine the relationship between quarterly serial dependence ρ_q and these explanatory variables x_q, we estimate the following regression for the U.S. market:

$$\rho_q = \alpha + \beta x_q + \gamma \rho_{q-1} + \epsilon_q. \tag{1}$$

To generalize our findings, we also conduct a pooled panel regression across multiple countries:

$$\rho_{i,q} = \alpha_i + \beta x_{i,q} + \gamma \rho_{i,q-1} + \epsilon_{i,q}, \tag{2}$$

where i indexes countries, and $\rho_{i,q}$ denotes the serial dependence of country i's index in quarter q. In the U.S. regression, all variables are standardized to have zero mean and unit standard deviation. In the global sample, standardization is applied within each country.[5]

Table 2 reports the regression results. Volatility consistently emerges as the most important and robust predictor of serial dependence. In the U.S. (Dow Jones Index), a one-standard-deviation increase in volatility reduces serial dependence by 3.21 percentage points (t-stat $= -4.56$). In the global sample, the same increase in volatility reduces serial dependence by 0.99 percentage points (t-stat $= -1.64$).

[5] The global sample includes the six non-U.S. markets examined above: the Tokyo Stock Price Index (Japan), CAC 40 (France), FTSE 100 (U.K.), S&P/TSX 60 (Canada), DAX (Germany), and S&P/ASX 200 (Australia). The Hang Seng Index is excluded due to missing term spread data from Bloomberg and concerns over its dual sensitivity to both mainland China macroeconomic conditions and Hong Kong monetary policy. Sample coverage for included markets begins around the 2000s, depending on data availability.

Table 2: The Impact of Market Variables on Index Serial Dependence. This table reports the results of time-series and pooled regressions based on equations (1) and (2). The dependent variable is the quarterly serial dependence of daily equity index returns, ρ_q, along with conditional measures based on the sign of daily returns: ρ_q^+ $(r_t \geq 0)$ and ρ_q^- $(r_t < 0)$, estimated from $r_{t+1} = a_q + \rho_q \cdot r_t + \epsilon_t$, for $t \in [q-1, q]$. The red line corresponds to ρ_q^+, and the blue line to ρ_q^-. The independent variables include the quarterly volatility of daily index returns (Vol_q), the quarterly average of monthly inflation (Inflation_q), the term spread (the yield difference between 10-year and 1-year Treasuries, $\text{Term}_q = y_q^{10\text{Y}} - y_q^{1\text{Y}}$), and the quarterly index return (Ret_q). For the U.S. market, additional independent variables include the short-term (3-month) Treasury yield ($r_q^{3\text{m}}$), the intermediary capital ratio proposed by He, Kelly and Manela (2017) (Capital Ratio_q), the quarterly average of monthly sentiment from Baker and Wrugler (2006), and NBER recession dummies (Recession_q). To address potential unit root issues, we control for lagged serial dependence (ρ_{q-1}). In U.S. market regressions, all independent variables are standardized to have a mean of 0 and a standard deviation of 1, while in global market regressions, standardization is applied within each country. The dependent variable is expressed in percentage points. The global index sample includes the Tokyo Stock Price Index (Japan), CAC 40 Index (France), FTSE 100 Index (U.K.), S&P/TSX 60 Index (Canada), DAX Index (Germany), and S&P/ASX 200 Index (Australia). The sample period spans from January 1897 to December 2024, with the starting date for each variable determined by data availability. For U.S. market regressions, t-statistics are based on Newey–West adjustments; for global market regressions, t-statistics are based on country–time two-way cluster adjustments.

	US Equity Index (Dow Jones)											Global Indexes		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)
yvar =	ρ_q									ρ_q^- $(r_t < 0)$	ρ_q^+ $(r_t \geq 0)$	ρ_q	ρ_q^- $(r_t < 0)$	ρ_q^+ $(r_t \geq 0)$
Vol_q	-3.21***								-4.08***	-7.12***	-2.79	-0.99	-4.24***	0.46
	[-4.56]								[-3.49]	[-4.35]	[-1.25]	[-1.64]	[-4.23]	[0.39]
Inflation_q		1.60**							1.72*	12.07***	-1.89	-0.05	0.61	-0.93
		[2.31]							[1.92]	[5.23]	[-1.00]	[-0.07]	[0.26]	[-0.92]
Term_q			-2.23***						-1.49*	5.35***	-3.33**	0.16	-0.34	0.12
			[-3.10]						[-1.89]	[2.81]	[-2.11]	[0.22]	[-0.28]	[0.13]
$r_q^{3\text{m}}$				1.10*										
				[1.70]										
Ret_q					0.66									
					[1.29]									
Capital Ratio_q						0.22								
						[0.31]								
Sentiment_q							-0.1							
							[-0.10]							
Recession_q								-0.7						
								[-0.47]						
ρ_{q-1}	0.33***	0.37***	0.39***	0.36***	0.39***	0.37***	0.42***	0.39***	0.31***	0.03	0.12**	0.09	-0.01	-0.01
	[6.91]	[6.32]	[7.50]	[5.78]	[8.42]	[4.56]	[6.07]	[8.36]	[4.96]	[0.54]	[2.04]	[1.29]	[-0.18]	[-0.43]
Intercept	2.07***	3.11***	3.33***	2.76***	1.88***	1.09	1.77**	1.98***	2.18***	-3.97**	5.68***		Country FE	
	[3.74]	[3.79]	[4.51]	[3.32]	[3.20]	[1.29]	[2.16]	[3.24]	[2.70]	[-2.03]	[3.28]			
NOBS	510	308	364	287	510	218	234	511	287	287	287	612	612	612
R2 (%)	20.59	16.06	16.65	18.1	15.85	14.09	17.77	15.31	23.82	11.39	4.21	7.05	8.43	3.51

Several macro variables – namely inflation and the term spread – also exhibit explanatory power in the U.S. when included individually. Lower inflation and a steeper yield curve are both associated with more negative serial dependence. However, when included alongside volatility, their significance disappears. Moreover, in the global regression, neither inflation nor the term spread shows significant explanatory power.

To better understand the strong role of volatility, we analyze its effect conditional on the sign of daily returns. For each quarter q, we separately compute serial dependence for negative and positive return days, denoted $\rho_q^-(r_t < 0)$ and $\rho_q^+(r_t \geq 0)$, respectively. If the impact of volatility comes through a liquidity channel, it should be more pronounced during market downturns – when liquidity providers face higher inventory risk and tighter capital constraints. Indeed, the empirical results support this hypothesis. In the U.S., a one-standard-deviation increase in volatility reduces serial dependence on negative return days by 7.12 percentage points (t-stat $= -4.35$), whereas the effect on positive return days is statistically insignificant. A similar pattern holds globally: volatility reduces negative-day serial dependence by 4.24 percentage points, while its effect on positive-day serial dependence is a modest and insignificant 0.46 percentage points.

In summary, volatility shows the strongest and most robust impact on equity index serial dependence, both in the U.S. and across international markets. Notably, this effect is also asymmetric: the pronounced return reversals associated with heightened volatility occur primarily following market declines.

3 Potential Mechanisms

Researchers have proposed several mechanisms to explain the presence of non-zero serial dependence in equity index returns. The substantial variation in serial dependence over time suggests that no single mechanism can fully account for its dynamics. While the overall level of serial dependence fluctuates considerably, certain robust empirical patterns – such as stronger return reversals following market declines, the disproportionate influence of extreme returns, and the pronounced role of volatility – offer valuable insights into the relative contribution of different explanatory channels. In this section, we review the main mechanisms proposed in the literature and evaluate their explanatory power in light of these empirical findings. We highlight both their strengths and limitations in accounting for the observed patterns of serial dependence in index returns.

Non-Synchronized Trading – Non-synchronous trading refers to the phenomenon where not all securities update their prices simultaneously due to infrequent trading. This lack of synchronization can induce positive serial dependence in aggregate index returns through the presence of stale prices. Previous studies by Fisher (1966), Atchison, Butler, and Simonds (1987), and Lo and MacKinlay (1988) emphasize non-synchronous trading as a key driver of observed positive autocorrelation in index returns. Further supporting this view, Ahn, Boudoukh, Richardson, and Whitelaw (2002) show that cash equity indices – which are more susceptible to non-synchronous pricing – exhibit greater serial dependence than their corresponding futures.

However, the relevance of this mechanism has likely declined in recent decades. The rise of electronic trading, increasing market liquidity, and the widespread adoption of high-frequency trading have made asset prices more synchronized over time. While this shift toward better synchronization may help explain the decline in positive serial dependence observed after the 1980s, it appears inconsistent with two key empirical facts: the emergence of negative serial dependence after 2000 and the presence of non-zero serial dependence before 1940, a period when trading was presumably even less synchronized. Moreover, the non-synchronous trading mechanism can not generate the asymmetric pattern in serial dependence following large market declines.

Sluggish Response to Information – Sluggish response, or investor underreaction to news, is a well-established mechanism contributing to positive serial dependence in returns. When markets adjust slowly to new information, returns tend to exhibit short-term persistence. A broad literature documents that stocks which incorporate information quickly tend to lead slower-reacting stocks, creating positive autocorrelation at the index level (Brennan et al. 1993; Badrinath et al. 1995; Chordia and Swaminathan 2000; Hou 2007; Moskowitz et al. 2012). This underreaction mechanism may have played a role in the gradual decline in serial dependence observed between the 1970s and 1990s, as improvements in information dissemination and trading speed increased investor responsiveness. However, this explanation falls short in explaining the increase in serial dependence after 1940 and the emergence of negative serial dependence post-2000. To reconcile these patterns, one would have to assume a behavioral shift from underreaction to overreaction. Moreover, explaining the asymmetry in serial dependence would require assuming that investors overreact to bad news while continuing to underreact to good news. Therefore, while sluggish response provides a plausible explanation for positive autocorrelation in certain periods, it does not fully capture the evolving and asymmetric nature of serial dependence across time.

Time-Varying Expected Returns – Another potential explanation is that index serial dependence arises from time-varying expected returns. Studies such as Fama and French (1988) and Conrad, Kaul, and Nimalendran (1991) demonstrate that changes in the expected return component of prices can generate return autocorrelation. Similarly, Chordia and Shivakumar (2002) link return predictability to lagged macroeconomic variables, suggesting that shifts in economic conditions may influence expected returns over time. However, applying this mechanism to daily return dynamics poses several challenges. Expected returns are generally thought to evolve at lower frequencies and are unlikely to exhibit significant variation on a day-to-day basis. The substantial swings in daily serial dependence that we document – both in magnitude and sign – would require unrealistic large and persistent fluctuations in daily expected returns to be consistent with this explanation.

Liquidity – The observed negative serial dependence in equity index returns strongly supports a liquidity-based explanation, wherein pronounced price reversals serve as compensation to liquidity providers for bearing inventory risk and funding constraints (Ho and Stoll 1981; Grossman and Miller 1988; Campbell, Grossman, and Wang 1993; Chordia, Roll, and Subrahmanyam 2002; Duffie 2010). Although this mechanism could not explain the strong positive serial dependence observed between the 1940s and 1990s, it is well aligned with the persistent asymmetric patterns documented across global markets – given that collateral and funding constraints typically become more binding following market downturns. Furthermore, the significant impact of volatility on serial dependence supports the liquidity-based explanation – elevated volatility increases the inventory risk faced by liquidity providers, thereby intensifying the need for compensation through price reversals.

4 Conclusion

We examine the evolution of serial dependence in U.S. equity index returns over the long period from 1897 to 2024. While theory predicts zero serial dependence under market efficiency, our analysis reveals substantial time variation and robust asymmetries. Serial dependence is near zero in the early 20th century, turns strongly positive after Great Depression, peaks at 1970s, and then becomes persistently negative after 2000. Despite shifts in level, a consistent pattern emerges: return reversals are significantly more likely following negative returns, particularly during extreme downturns. This asymmetric behavior is robust across major international equity markets. Among potential drivers, volatility – especially during market declines – emerges as the strongest and most robust determinant of negative serial dependence in equity index returns, supporting liquidity-based explanations. In contrast, traditional channels such as non-synchronous trading and sluggish price adjustment account for only part of the observed dynamics.

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